Product Catalogue Application – Cloud Computing Assignment 2

Overview

This project implements a Product Catalogue Web Application deployed on a Public Cloud platform (Microsoft Azure). The application allows users to add new products and view a list of all products through a simple web interface.

The application is hosted on Azure App Service, which provides a publicly accessible URL and scalable cloud infrastructure.

Public URL of the application:

https://productcatalogcloudcomputingassn2.azurewebsites.net

Features Added and Their Usefulness

Azure App Service (Cloud Hosting)

The application is deployed using Microsoft Azure App Service, which provides a managed cloud platform for hosting web applications.

Why it is useful:

- Provides a **publicly accessible URL**
- Handles **deployment and hosting automatically**
- Allows applications to **scale easily**

This enables the product catalogue application to be accessed from anywhere through the internet.

Azure Monitoring and Logging

Azure App Service provides built-in application monitoring and logging capabilities.

Why it is useful:

- **Tracks application performance**
- Helps **detect runtime errors**
- Enables **debugging and monitoring** of application behavior

This helps ensure reliability and easier maintenance of the application.

Architecture Decisions

Decision	Alternatives	Choice	Justification

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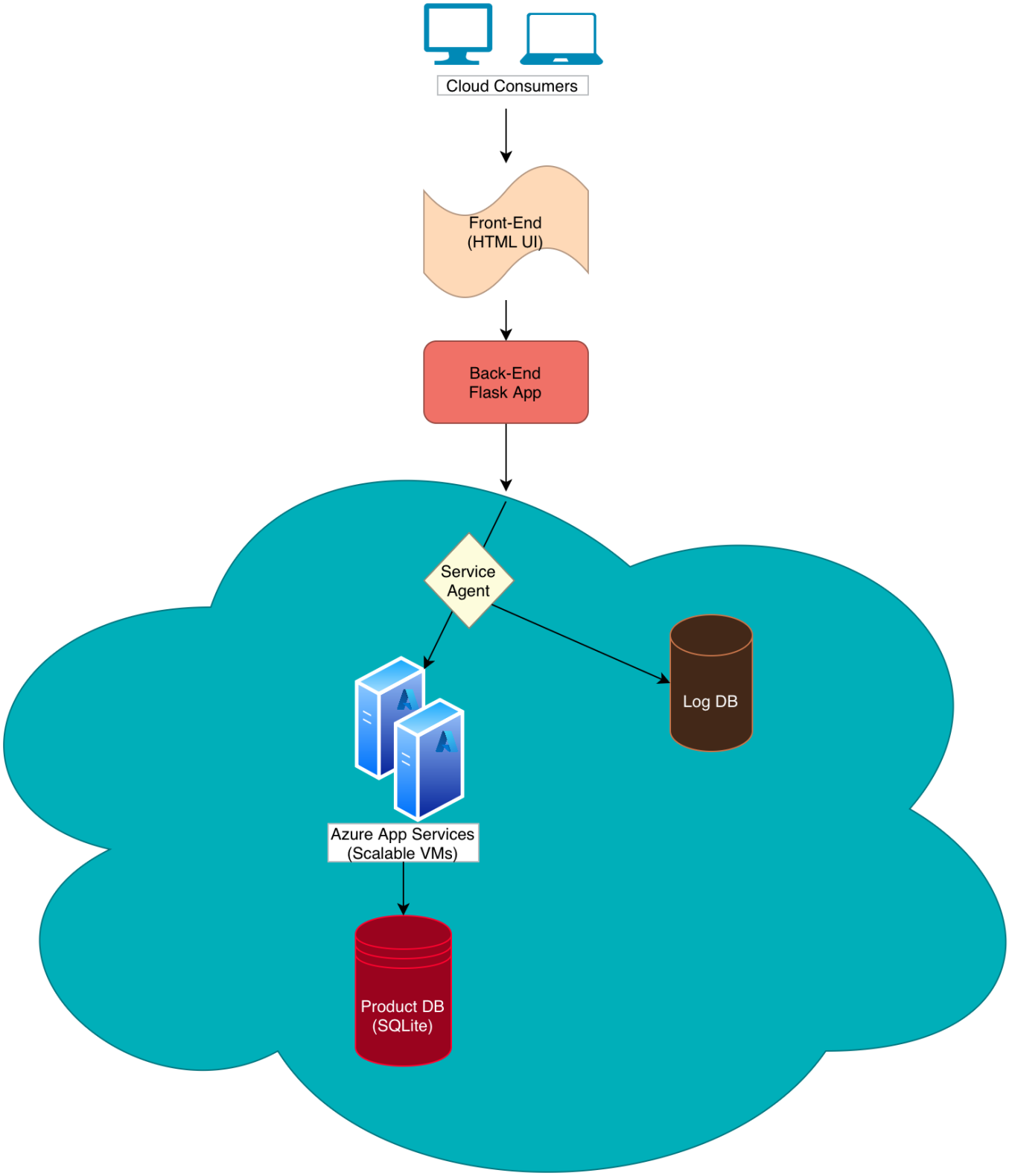
Web Framework	Django, Node.js (Express), Flask	Flask	Lightweight and easy to develop with, requires minimal configuration, suitable for small web applications and prototypes, and easy to deploy on cloud platforms.
Database	MySQL, PostgreSQL, NoSQL (MongoDB), SQLite	SQLite	Lightweight and serverless database, simple integration with Python, suitable for small applications with moderate data requirements, and requires no additional database server configuration.
Cloud Platform	AWS, Google Cloud Platform (GCP), Microsoft Azure	Microsoft Azure	Provides fully managed cloud hosting through Azure App Service, offers built-in monitoring and logging tools, and allows quick deployment without complex infrastructure configuration.

Architecture Diagram


Cloud Consumers
Front-End
(HTML UI)
Back-End
Flask App
Service
Agent
Log DB
Azure App Services
(Scalable VMs)
Product DB
(SQLite)